

ECON 0100 | Vignette B4 | Efficiency

Due in Recitation. Vignettes are a certificate of the work done together in Recitation.

The supply and demand curves for food before the pandemic are given in **Vignette B3**:

$$S : P_s = 2 + \frac{1}{2}Q_s \qquad D : P_d = 20 - Q_d$$

The equilibrium was 12 daily servings at 8 dollars.

Q1 | Supply Chain Shock

The pandemic substantially disrupted the supply chain, leading to difficulty bringing food to market with a new supply curve below.

$$P = 5 + \frac{1}{2}Q_s$$

Use the graph (*above*) to discuss the impact of supply chain disruptions on the market. Be sure to include how welfare measures have changed.

- a) New equilibrium price: _____ quantity: _____
- b) Consumer surplus: _____ Producer surplus: _____
- c) Change in total surplus: _____

Q2 | A Ban on Price Gouging

In response to worries about price-gouging a basic necessity during a time of difficulty, the government imposes a ban on price hikes from pre-pandemic levels. Use a graph to discuss the impact on food availability were this policy to be implemented. Show welfare measures and calculate the new consumer surplus.

a) Quantity supplied: _____ Quantity demanded: _____ Shortage:

b) Consumer surplus: _____ Producer surplus: _____ Deadweight loss:

c) Compared with the market after the shock, buyers: *gain, lose* and sellers: *gain, lose*